# Digital logic

National 5 · Electronics & control · Worksheet

ENGINEERING WITH MR KELLY

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ complete truth tables for AND, OR and NOT and combinations of gates
- ☐ turn a written problem into a logic circuit

## I DO — Worked example

Read how a full-mark answer is built

*A machine guard system: the motor runs (Z) only when the guard is closed (A) AND the start button is pressed (B), OR when the maintenance key (C) is on.*

**1 Draw the logic diagram.** (3 marks)

**Points that earn marks** from the marking instructions: one ✓ per point

- ✓ AND gate with inputs A and B
- ✓ OR gate with the AND output and C
- ✓ Output Z from the OR gate

## WE DO — Guided practice

Use the starters and word bank to help you

*A circuit has inputs A and B into an OR gate whose output goes through a NOT gate to Z.*

**2 Complete the truth table for Z.** (2 marks)

**Sentence starters**

One … is …, which means …

Another … is …, so …

**Word bank**

NOR

Tip: 2 marks = 2 separate, developed points.

## YOU DO — On your own

No notes. Check your answers afterwards.

**3 Quick check: tick one box for each.**

**a)** A=1, B=0 into an AND gate gives…

☐ 0 ☐ 1 ☐ Both ☐ Neither

**b)** A=1, B=0 into an OR gate gives…

☐ 0 ☐ 1 ☐ Undefined ☐ 2

**c)** How many rows does a 3-input truth table have?

☐ 3 ☐ 6 ☐ 8 ☐ 9

**4 Key terms: write the letter of the matching description next to each term.**

| | | |
|---|---|---|
| 1. **AND gate** | ____ | **A** Output 1 only when all inputs are 1. |
| 2. **OR gate** | ____ | **B** Output is the opposite of the input. |
| 3. **NOT gate** | ____ | **C** Output 1 when any input is 1. |

**5 Explain what is meant by ‘Rows in a 3-input truth table’.** (2 marks)

**Check and reflect**

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?